AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 23 · STUDENT LESSON

Deployment environments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Queues and asynchronous work. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of deployment environments. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of deployment **environment**s.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of deployment **environment**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Queues and asynchronous work. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain deployment **environment**s using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Deployment environments case. Separate observations, given facts and assumptions before choosing an explanation.

Development, staging and production are distinct **environment**s with explicit data and credential boundaries. Test configuration must not silently target production.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Deployment environments. Connect each labelled input, deciding step and result.

For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed **environment** names. A staging run cannot use production write credentials.

Deployment environments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Staging configuration points to production storage while the interface displays staging.	For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names. A staging run cannot use production write credentials.	The label is misleading; block the run through configuration validation. A display badge is not an isolation boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Deployment environments. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Point staging at the production credential.

Predicted result

Prediction: Stop the rehearsal because its separation boundary is breached.

Why the result follows

Reason: Actual access matters more than the **environment** label.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Deployment environments to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Environment	A declared deployment/configuration context
Isolation	Preventing unintended cross-context access
Configuration mapping	The relationship between environment and allowed resources

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Deployment environments, predict a step, then compare with its explanation.

Staging configuration points to production storage while the interface displays staging.

The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Worked explanation · Deployment environments

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Deployment environments

Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.

The facts we will use

Supplied facts

Staging card names trial room Z; production card names room A. Credentials and stores are separate.

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Deployment environments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Use staging data

The rehearsal reads fictional room Z.



2. Check actual connection

Staging credentials must not reach production storage.



3. Review evidence

A passing rehearsal supports release review, not automatic promotion.

CHANGE ONE THING

Point staging at the production credential.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Use staging data	The rehearsal reads fictional room Z.
2. Check actual connection	Staging credentials must not reach production storage.
3. Review evidence	A passing rehearsal supports release review, not automatic promotion.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An **environment** includes code, configuration, permissions and data. A staging label alone cannot create separation.

What this example does not establish

Keep the stated limit

Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

Keep the stated limit

Read the labelled picture: Identify the staging data and credential boundary. Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

Keep the stated limit

Change one thing: Point staging at the production credential. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Deployment environments claim and explain a repair using the supplied evidence.

Passing staging tests does not prove production limits or operational conditions are identical.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Passing staging tests does not prove production limits or operational conditions are identical.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Deployment environments with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Deployment **environment**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Deployment **environment**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Identify the staging data and credential boundary. Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Deployment environments. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Deployment environments. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Passing staging tests does not prove production limits or operational conditions are identical.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled deployment **environment**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of deployment **environment**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Deployment environments case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Deployment environments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Staging configuration points to production storage while the interface displays staging.	For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names. A staging run cannot use production write credentials.	The label is misleading; block the run through configuration validation. A display badge is not an isolation boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

> Development uses fake records but a production token. What boundary is violated?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

> Development uses fake records but a production token. What boundary is violated?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Deployment environments, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.

By the end, I can	How I will show it
Explain the mechanism and terminology of deployment environments	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Deployment environments without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of deployment **environment**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A passing rehearsal supports release review, not automatic promotion.

1. Use staging data

2. Check actual connection

3. Review **evidence**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Review **evidence** A passing rehearsal supports release review, not automatic promotion. An **environment** includes code, configuration, permissions and data.

A staging label alone cannot create separation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Point staging at the production credential. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Stop the rehearsal because its separation boundary is breached.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop the rehearsal because its separation boundary is breached. Actual access matters more than the **environment** label.

Repair a missing building block

Review Queues and asynchronous work (ai-6/22). Goal: Explain the **mechanism** and terminology of queues and asynchronous work. Short practice: Explain what the queued J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?. Return to this week and explain what you changed.

Review Data retention and deletion (ai-6/20). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Apply the seven-day boundary to T1 and T2.

How many T2 locations need verification, and may deletion be called complete if its backup remains?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Keep your completed work

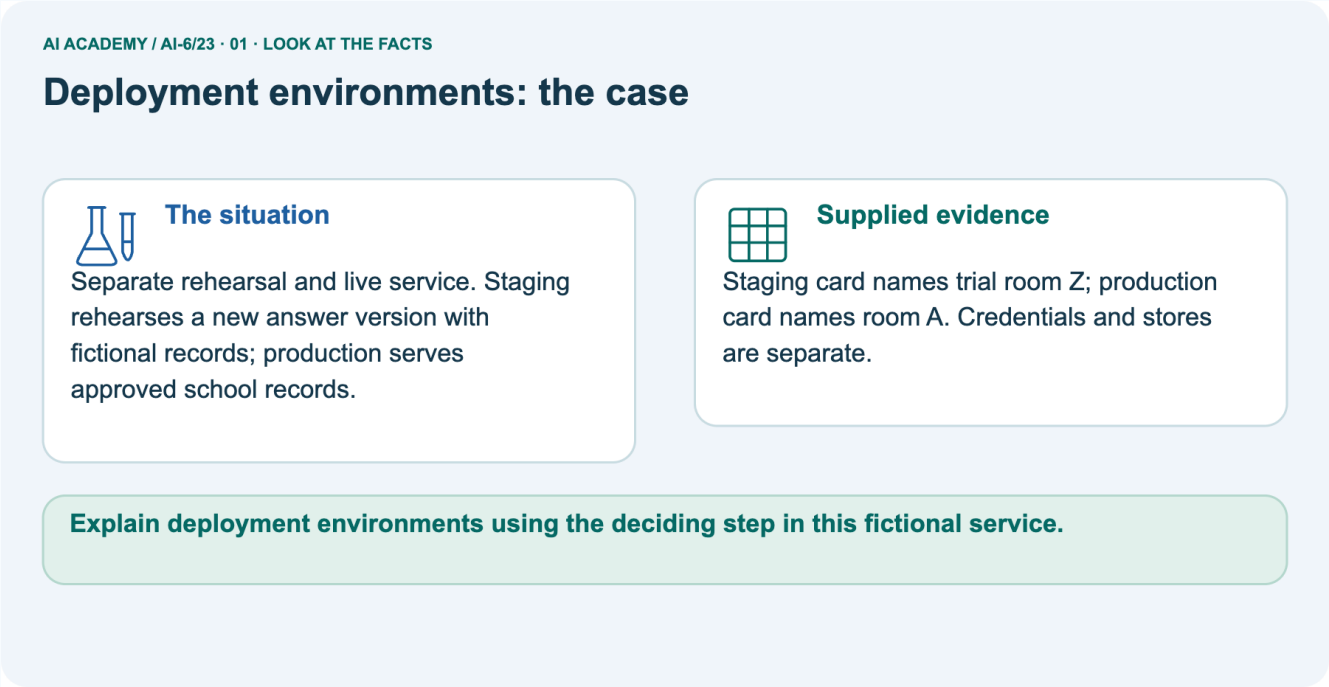
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Deployment environments graphic using its labels, deciding evidence and relevant limitation.

01. Deployment environments: the case



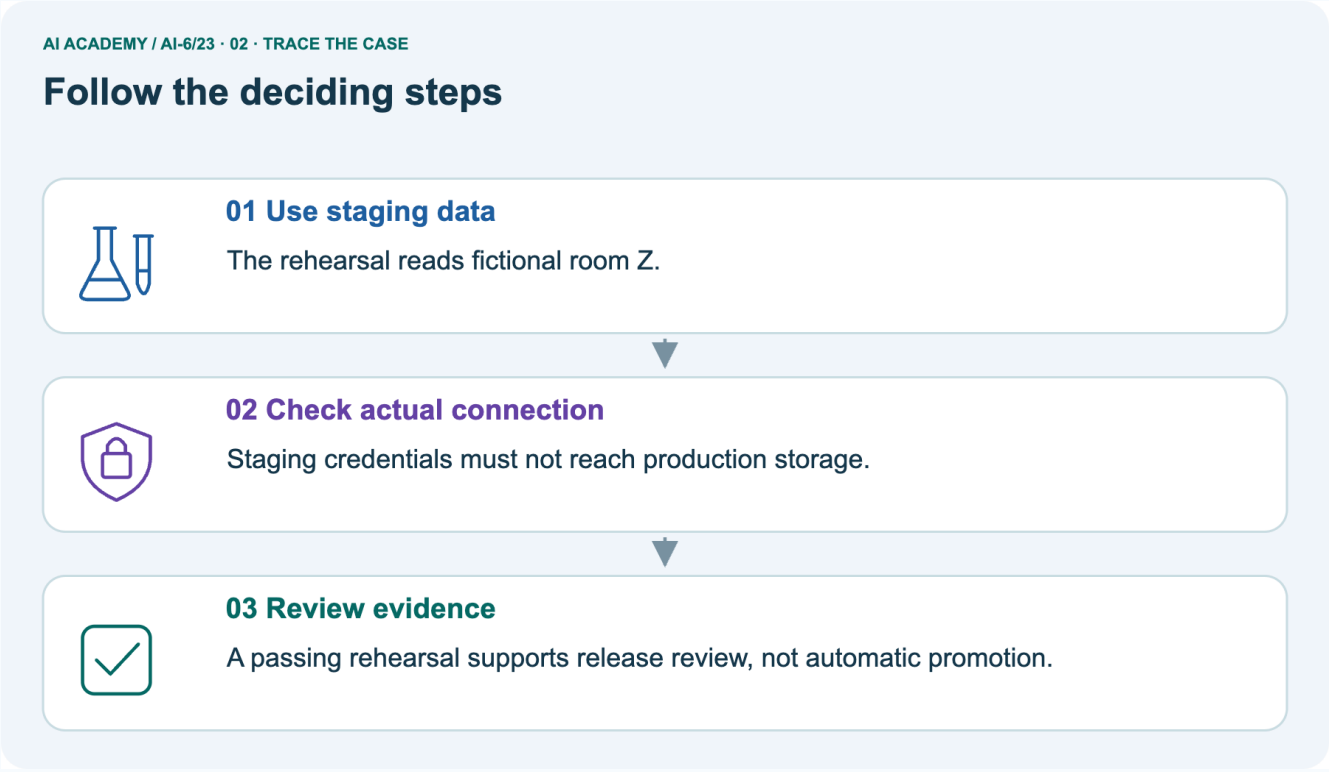
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records. Supplied facts: Staging card names trial room Z; production card names room A.

Credentials and stores are separate.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

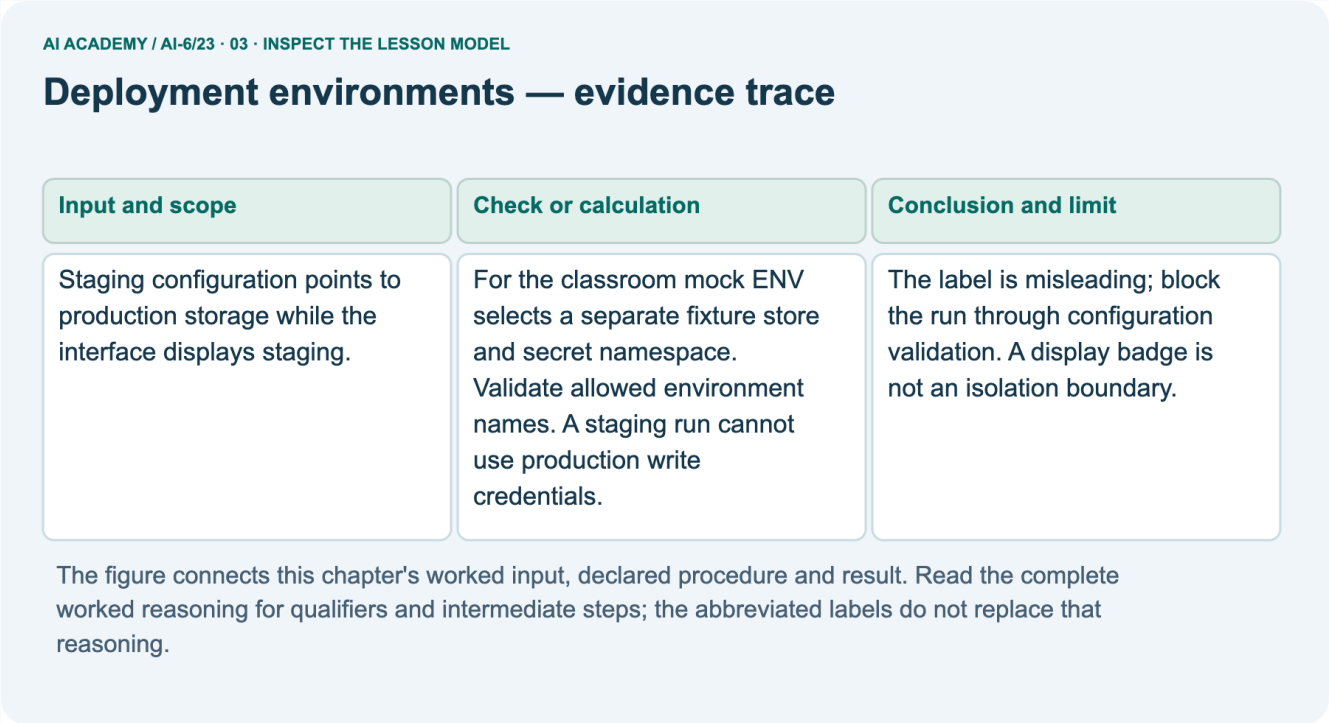


Follow the labelled picture with the matching explanation. Open the vector diagram

Use staging data: The rehearsal reads fictional room Z. → Check actual connection: Staging credentials must not reach production storage. → Review **evidence**: A passing rehearsal supports release review, not automatic promotion.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Deployment environments — evidence trace



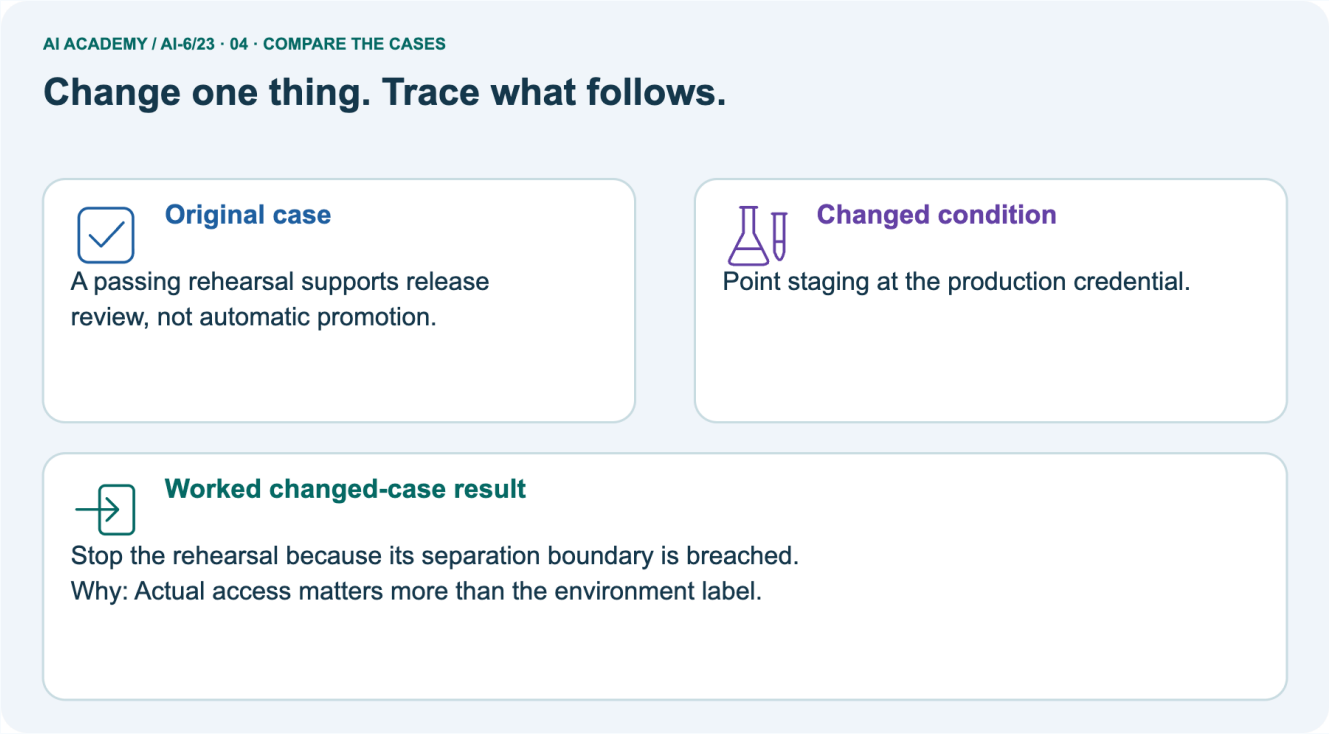
Follow the labelled picture with the matching explanation. Open the vector diagram

Deployment **environment**s — **evidence** trace. Staging configuration points to production storage while the interface displays staging. | For the classroom mock ENV selects a separate fixture store and secret namespace.

Validate allowed **environment** names. A staging run cannot use production write credentials. | The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



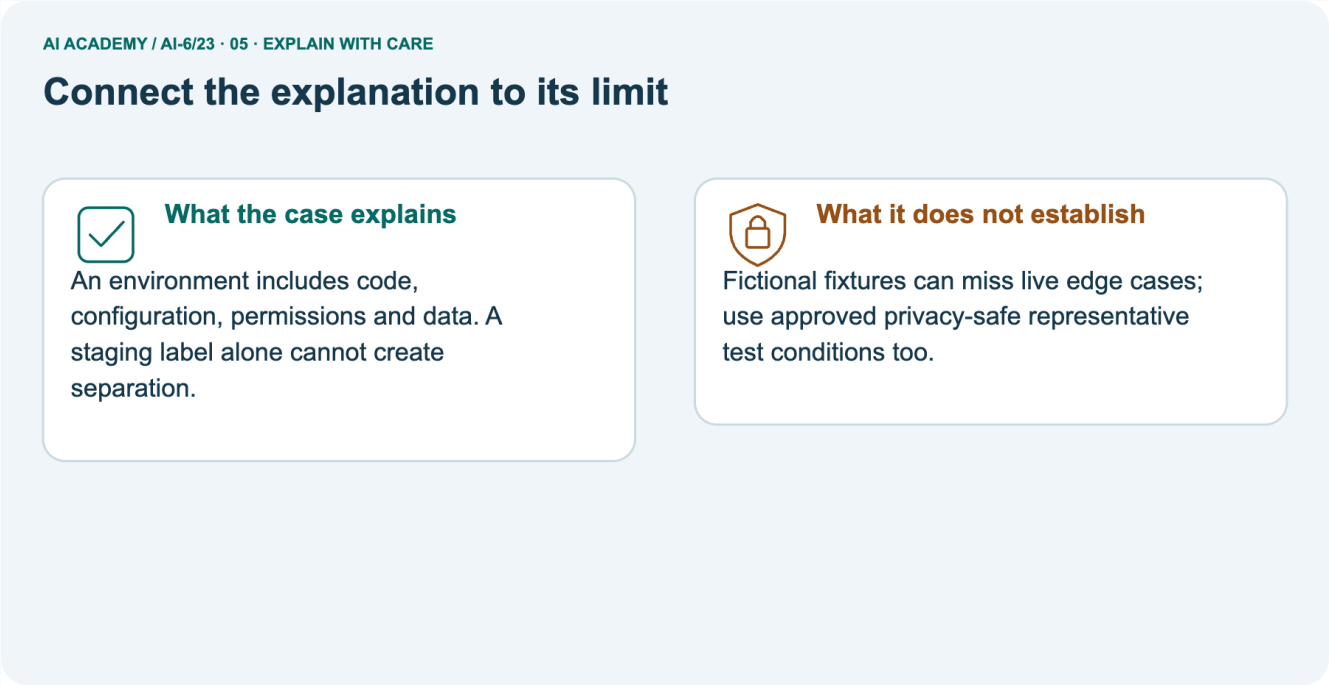
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A passing rehearsal supports release review, not automatic promotion. Changed condition: Point staging at the production credential.

Expected result: Stop the rehearsal because its separation boundary is breached. Reason: Actual access matters more than the **environment** label.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



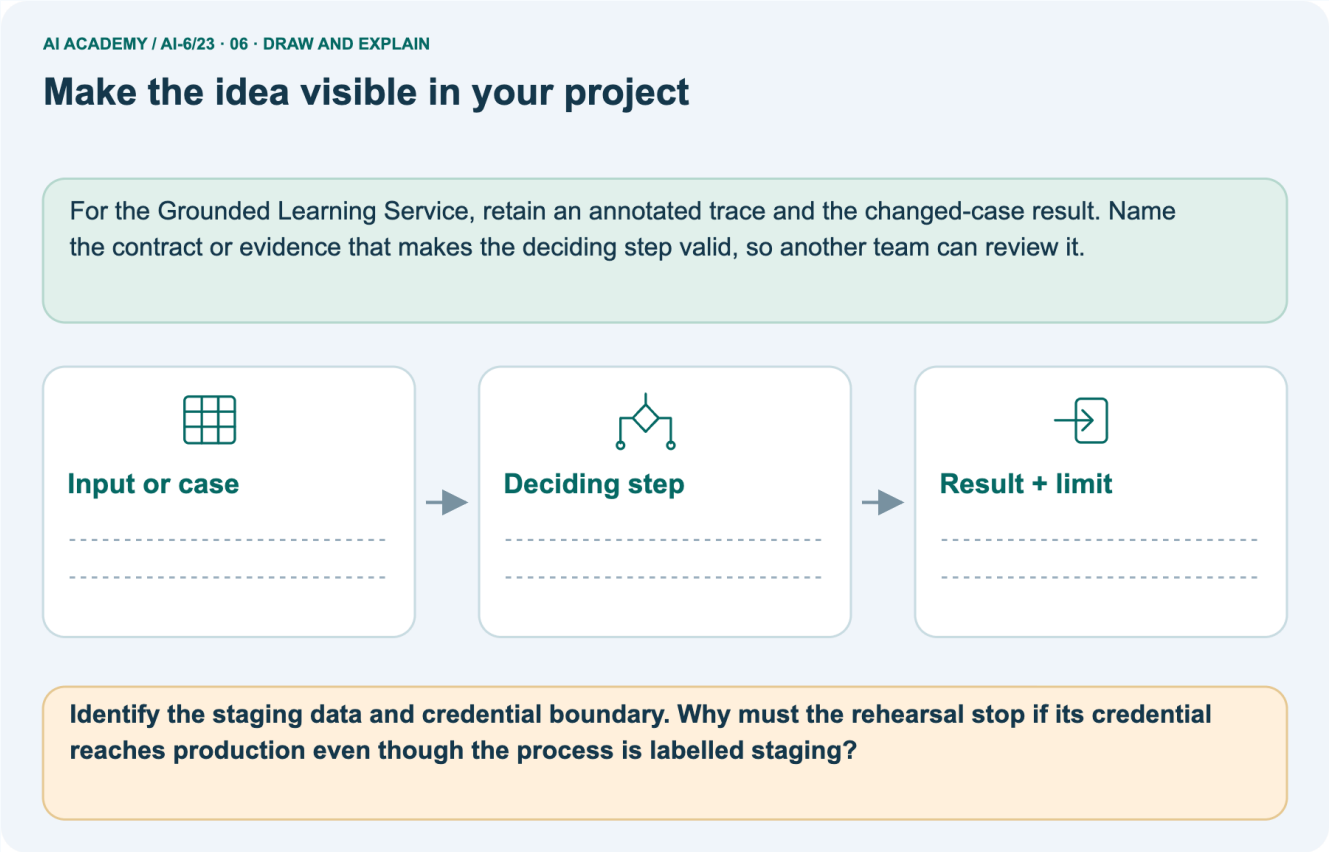
Follow the labelled picture with the matching explanation. Open the vector diagram

An **environment** includes code, configuration, permissions and data. A staging label alone cannot create separation.

Limit: Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.